

Christopher Eder CHANTRES JUSTO

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EDUCATION

Universidad de las Américas Puebla (UDLAP) - Mexico

August 2024 – May 2028

- Bachelor of Science in Data Science | GPA 3.9/4.0
 - Algorithms – Calculus 2 - Stochastic Models – Matrix Theory – OOP – Data Structures – Databases
- Member of the Honours Program - Scientific Research on Graph Theory and its Applications. **June 2025 - Present**
- Co-Founder & President at Bitwise Competitive Programming Club **Feb 2025 - Present**
 - Led competitive programming & career growth workshops for +30 students on weekly sessions.
- STEM Ambassador **July 2025 - Present**
 - Mentored +85 high school students during UDLAP's STEM Summer Week “Internship Preparation” workshops.
 - Served as a speaker, competitor, and team leader for UDLAP's official STEM representative team.

EXPERIENCE & COMPETITIONS

Software Engineer (Part Time) | Universidad de las Américas Puebla

August 2025 – Feb 2025

- Engineered a shared FastAPI backend to serve business logic for separate iOS, Android, and Web clients.
- Architected a robust, type-safe QR access control microservice by translating user personas and requirements into design.
- Built repository layer with full CRUD, pagination, and filter-based search (location, date range, IDs).

Software Engineer Fellow | Uber

December 2024 – August 2025

- Accepted from the top 3% of a global applicant pool to join the Uber Career Prep (UCP) 2025 Undergraduate Cohort.
- Analyzed and solved problems through advanced data structures and algorithms.
- Engaged in workshops and mentorships to strengthen technical skills, interview readiness, and advance career growth.

2X International Collegiate Programming Contest (ICPC) Mexican Finals 2024 & 2025

November 2024 & 2025

- Qualified for the ICPC Mexican Finals by solving problems and optimizing solutions in C++ under time constraints.
- Ranked 63rd out of 120 teams in my first ICPC Mexico Finals, showcasing problem-solving skills and resilience.
- Obtained an “Honorable Mention” thanks to my commitment and team support throughout the competition.

National iOS Development Lab Hackathon Swift Change Makers | Apple & Enactus Mexico

May 2025

- Secured 2nd place in the category of “Best Solution” by gamifying the process of creating business bank accounts.
- Engineered an MVP in Swift for BBVA that automated SME legal registration, serving a market of 3.2 million businesses.
- Directed UI/UX design in Figma (wireframes & aesthetics) and crafted the project pitch.

PROJECTS

Server Capacity Forecaster | Machine Learning

January 2026

- Researched and implemented a Linear Regression based on Ordinary Least Squares (OLS) from scratch using NumPy.
- Designed a Dockerized full-stack interactive application that predicts server CPU usage based on simulated user traffic.

Atomic-Vec | Database Systems, Concurrency

January 2026

- Engineered a thread-safe, snapshot-isolated vector database engine from scratch in C++20, utilizing Multi-Version Concurrency Control (MVCC) to support simultaneous read/write operations.
- Architected a Transaction Manager governed by a logical clock system, with lock-free reads and consistent data snapshots.
- Implemented automated write conflict detection and strict visibility rules with transaction rollbacks.

Sargassum Classifier AI Model | Computer Vision & Deep Learning

June 2025

- Developed a multi-class image classification model using PyTorch and Swin Transformer architecture to predict Sargassum seaweed levels in Mexican Caribbean beaches, achieving 4th place in MEIA UNAM 2025 competition.
- Implemented ML techniques including class imbalance handling, data augmentation, and early stopping with learning rate scheduling to optimize model performance.
- Built deep learning pipeline with transfer learning and comprehensive evaluation metrics (precision, recall, F1-score).

SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, C/C++, Java, Swift.
- **Technologies:** Docker, Git, GitHub, Linux, PostgreSQL, HTML, CSS, Arduino UNO, Figma.
- **Frameworks/Libraries:** FastAPI, Django, React.js, Next.js, Pandas, PyTorch.
- **Spoken Languages:** Spanish (Native), English (B2 DET), French (A2).